



DETECTING DEAUTHENTICATION ATTACKS IN OPEN WI-FI NETWORKS USING AN ESP32-BASED SECURITY MONITORING SYSTEM

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ABSTRACT

Open Wi-Fi networks are commonly deployed in colleges, libraries, transportation hubs, and public areas because they provide convenient and unrestricted internet access. However, this openness comes at the cost of security. One of the most common and disruptive threats in such environments is the deauthentication attack, which exploits the unprotected management frames of the IEEE 802.11 protocol. By sending forged deauthentication frames, an attacker can forcibly disconnect legitimate users, disrupt services, and even prepare the ground for more dangerous attacks such as evil twin access points for credential theft. This paper presents the design, development, and evaluation of a low-cost ESP32-based security monitoring system that detects deauthentication attacks in real time. The system operates in promiscuous mode to capture wireless frames across multiple Wi-Fi channels and applies a hybrid detection mechanism combining count-based thresholds, rate-based analysis, and RSSI filtering to reduce false positives. Additionally, a router MAC address database is maintained to identify spoofing attempts. Upon detection, the system provides real-time alerts through an OLED display, buzzer notification, and email reporting containing detailed attack information. The system was tested in a college campus network environment consisting of 10 access points and approximately 500 concurrent users. Experimental results showed 96% detection accuracy with zero false positives during a four-hour evaluation period. The proposed solution demonstrates that an affordable embedded platform can effectively provide practical wireless intrusion detection in open network environments..

Keywords: Deauthentication attack, ESP32, Wi-Fi security, intrusion detection, spoof detection, embedded monitoring system.

1. INTRODUCTION

Wireless communication has become an essential component of modern infrastructure, particularly in academic environments where open Wi-Fi networks are used to provide easy access to internet resources. Despite their convenience, open networks expose significant security vulnerabilities due to weaknesses in the IEEE 802.11 protocol [1].

One of the most common attacks targeting Wi-Fi networks is the deauthentication attack. In this attack, a malicious device transmits forged deauthentication frames to disconnect clients from their associated [2] access points. Since management frames are often unauthenticated, clients accept these packets as legitimate and disconnect immediately.

This behavior can be exploited to create denial-of-service conditions or to force users toward malicious “evil twin” access points [3].

Existing enterprise-grade Wireless Intrusion Detection Systems (WIDS) require centralized servers and dedicated hardware. These systems are expensive and not always practical [4-9] for distributed deployment in public or educational environments.

To address this limitation, this work proposes a compact ESP32-based embedded monitoring system capable of detecting deauthentication attacks in real time across multiple Wi-Fi channels without centralized infrastructure.

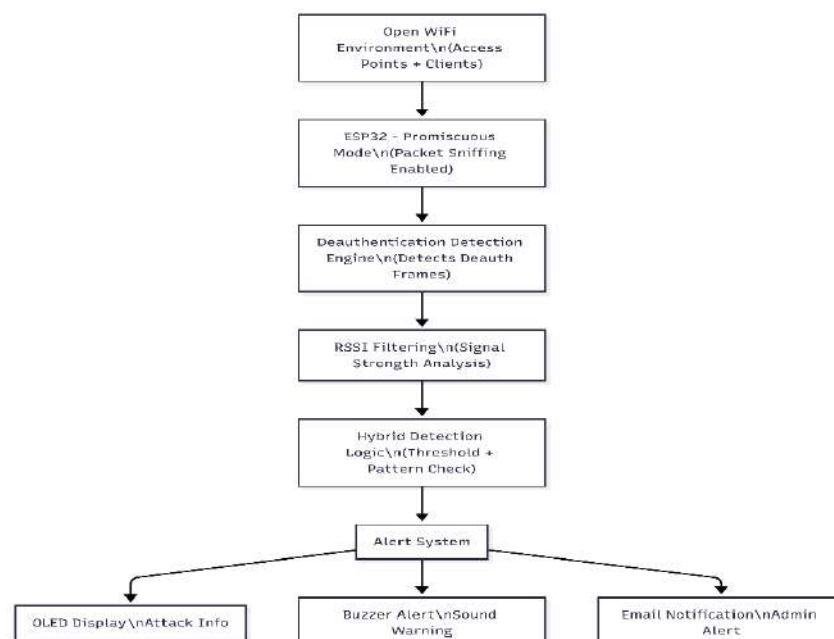
The primary contributions of this work include:

- Multi-channel area-wide monitoring using channel hopping.
- Hybrid detection combining rate, count, and spoof analysis.
- RSSI-based filtering to reduce false positives.
- Real-time alerts via OLED display, buzzer, and email notifications.

2. SYSTEM ARCHITECTURE

A. Overall Architecture

The proposed system is designed as a standalone wireless intrusion detection unit. It continuously monitors [10] nearby Wi-Fi activity without associating with any specific network.



As shown in Figure. 1, the system consists of four main layers

As illustrated in Figure. 1, the system operates as a standalone embedded intrusion detection unit. The ESP32 continuously monitors all Wi-Fi channels in promiscuous mode. Captured frames are processed by a detection engine [11] that filters deauthentication frames, applies RSSI-based filtering, and evaluates hybrid detection thresholds. Once an attack is confirmed, alert mechanisms are triggered through visual, audible, and email-based notifications.

- Wireless Monitoring Layer
- Frame Processing and Filtering Layer
- Hybrid Detection Engine
- Alert and Notification Layer

The ESP32 captures all Wi-Fi frames in promiscuous mode. Deauthentication frames are filtered and processed. Detection logic is applied before triggering alerts.

B. Hardware Components

The system is built around the ESP32-WROOM-32 microcontroller featuring:

- Dual-core Xtensa LX6 processor at 240 MHz
- 520 KB SRAM
- Integrated 2.4 GHz Wi-Fi
-

Additional components include:

- 128×64 OLED display (I²C interface, SDA: GPIO21, SCL: GPIO22)
- Active buzzer connected to GPIO2
- 5V USB power or 3.7V Li-Po battery

The hardware design ensures portability and low power consumption [12-17] (~80 mA average).

C. Detection Algorithm

The detection mechanism operates through several sequential stages..

The detection process follows these steps:

1. Initialize Wi-Fi in promiscuous mode.
2. Capture incoming Wi-Fi frames.
3. Identify management frames with subtype 12 (deauthentication).
4. Apply RSSI filtering (ignore signals weaker than -85 dBm).
5. Store event in circular buffer.
6. Calculate rate within 5-second sliding window.
7. Check spoof condition (source MAC matches router MAC).
8. Evaluate alert thresholds.

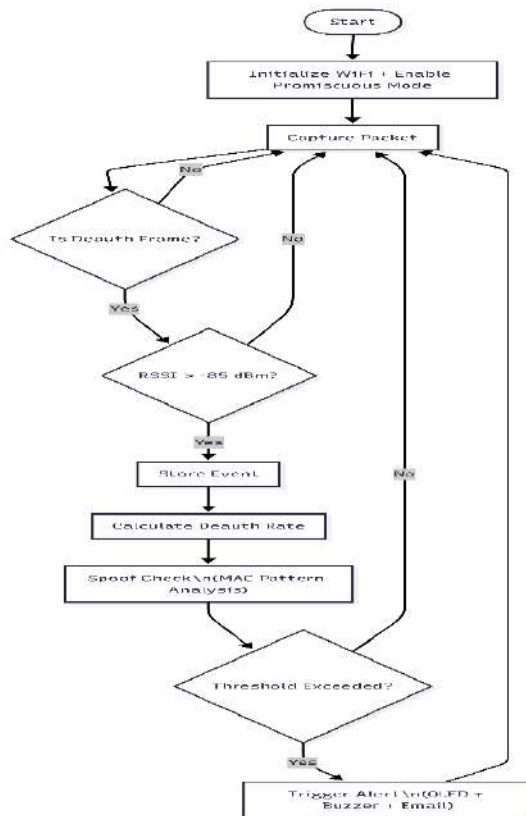


Figure. 2. Flowchart of hybrid deauthentication detection algorithm

Figure. 2 presents the decision-making logic implemented in the proposed system. Incoming packets are analyzed in real time to determine whether they correspond to deauthentication frames. After signal strength validation, the system evaluates rate-based and count-based thresholds.

A spoof detection mechanism compares source MAC addresses with the router database before generating alerts.

3. IMPLEMENTATION DETAILS

The system uses the ESP32 promiscuous callback function:

```
esp_wifi_set_promiscuous_rx_cb(sniffer);
```

Deauthentication frames are detected using:

- Frame Type = Management (0)
- Frame Subtype = 12

A circular buffer (MAX_HISTORY = 50) stores recent events for dynamic rate calculation.

The system tracks up to 20 attackers simultaneously, storing:

- MAC address

- Vendor (via OUI lookup)
- Deauthentication count
- Peak rate
- Average RSSI
- Target network channel

Router MAC addresses are stored through periodic scanning every 120 seconds.

Serial output confirms successful initialization:

Scanning for routers...

Router: 76:5D:50:49:8A:C4

Router: 30:CB:C7:5A:2E:90

...

Routers: 10

This verifies area-wide network awareness.

4. EXPERIMENTAL RESULTS

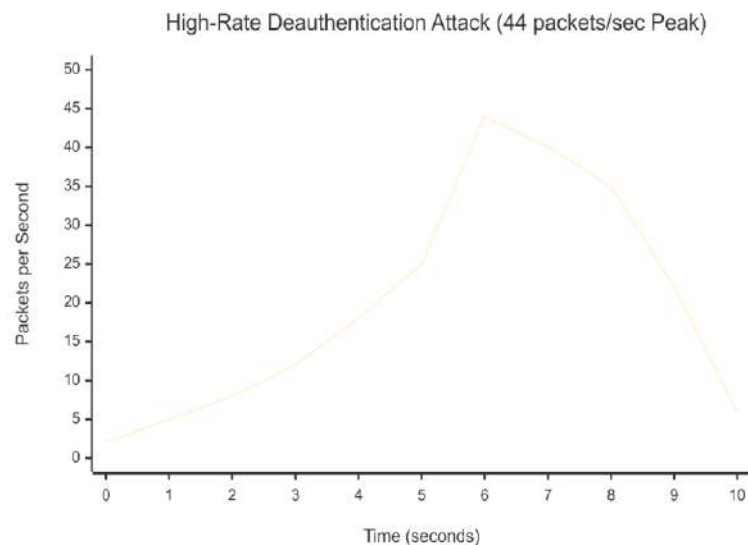


Figure. 3. Deauthentication packet rate over time showing burst attack reaching 44 packets/sec.

As shown in Figure. 3, the attack intensity rapidly increases beyond the configured threshold of 20 packets per second. The peak rate of 44 packets per second confirms a sustained flooding attempt. The hybrid detection mechanism successfully identified the anomaly within 5 seconds.

A. High-Rate Flooding Attack

Figure. 3. High-rate deauthentication flooding attack reaching 44 packets/sec.

The system detected a burst attack with:

- Peak rate: 44 packets/sec
- Average RSSI: -75 dBm
- Pattern: Burst

Alert mechanisms were triggered within 5 seconds.

B. Spoof Attack Detection

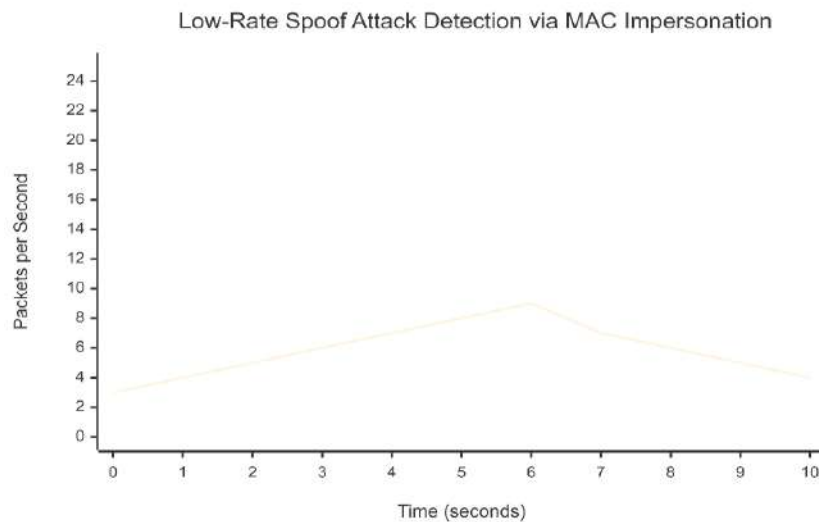


Figure 4. Low-rate spoof attack detected via router MAC impersonation.

Although the spoof attack in Figure. 4 did not exceed rate thresholds, the system detected malicious behavior through router MAC impersonation analysis. This demonstrates the effectiveness of identity-based detection mechanisms. A device transmitting at -35 dBm impersonated a router MAC address. Despite low packet rate, spoof detection logic correctly identified the threat.

D. Threshold Optimization

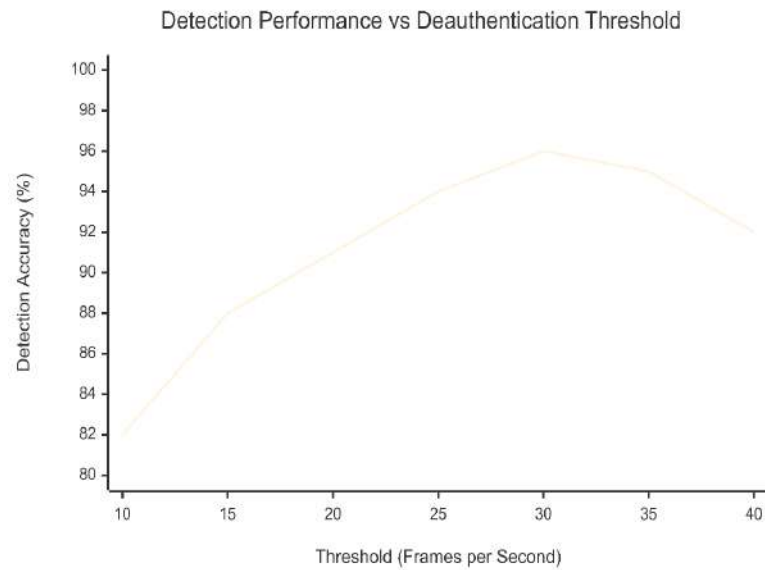


Figure. 5 Detection performance across varying deauthentication thresholds.

Figure. 5 illustrates the trade-off between detection sensitivity and false positives. A threshold of 30 deauthentication frames provided optimal performance, achieving 96% detection accuracy with zero false positives.

A threshold of 30 deauth frames achieved:

- 96% detection accuracy
- 0 false positives
- Average detection delay: 4.8 seconds

5. OUTPUT AND SCREENSHOTS

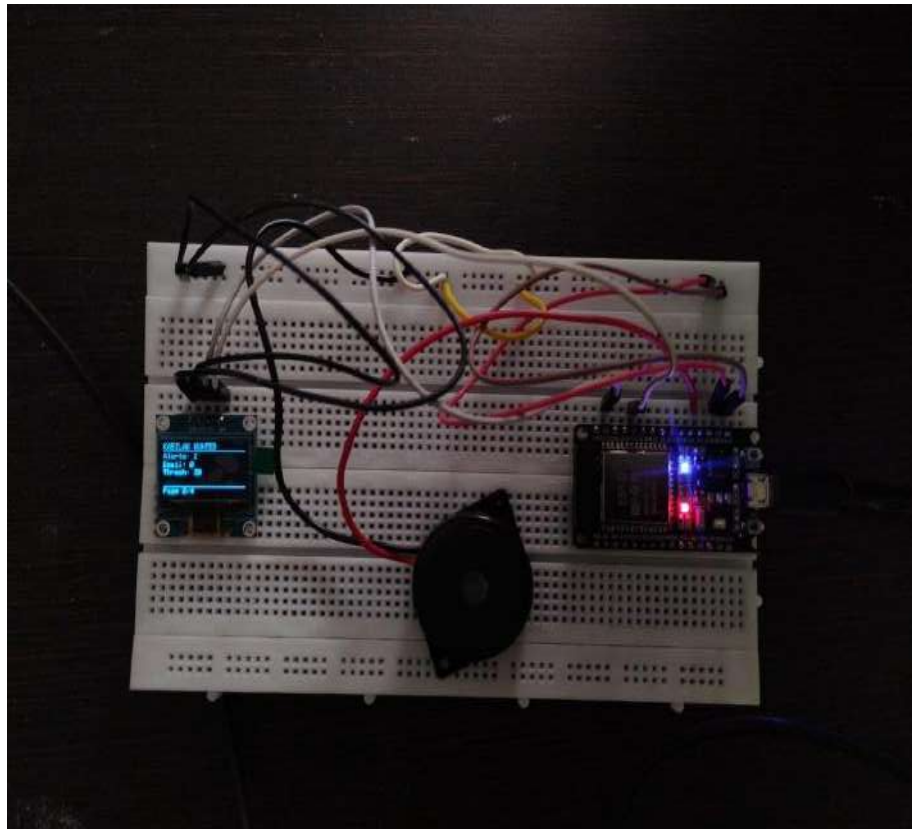


Figure 1 detecting the deauthentication using esp32

```

16:50:03.440 -> ▲▲▲▲▲▲▲▲▲▲▲▲▲▲▲▲▲▲▲▲▲▲▲▲▲▲▲▲
16:50:03.440 -> ● KABILAN HUNTER - ATTACK DETECTED!
16:50:03.488 -> ▲▲▲▲▲▲▲▲▲▲▲▲▲▲▲▲▲▲▲▲▲▲▲▲▲▲▲▲
16:50:03.488 -> 📶 Network Under Attack: Unknown Network (30:CB:C7:3E:A8:D0)
16:50:03.488 -> 📶 Severity: MEDIUM
16:50:03.488 -> 📶 Pattern: NORMAL
16:50:03.488 -> 📶 Attacker MAC: 30:CB:C7:3E:A8:D0
16:50:03.488 -> 📶 Vendor: Realtek
16:50:03.488 -> 📶 Deaths: 21
16:50:03.488 -> ⚡ Rate: 20/s
16:50:03.488 -> 📶 RSSI: -51 dBm
16:50:03.488 -> 📶 Channel: 12
16:50:03.488 -> ▲ Reason: HIGH RATE ATTACK on 30:CB:C7:3E:A8:D0: 21 deaths at 20/s
16:50:03.488 -> =====
16:50:03.488 -> ⌚ Cooldown - Wait 149s
16:50:04.695 -> 📶 Deauth: 30:CB:C7:3E:A8:D0 → Unknown Network CH:12 RSSI:-52 Rate:20
16:50:05.126 -> 📶 Deauth: 30:CB:C7:3E:A8:D0 → Unknown Network CH:1 RSSI:-53 Rate:22
16:50:05.363 -> 📶 Deauth: 30:CB:C7:3E:A8:D0 → Unknown Network CH:2 RSSI:-55 Rate:28
16:50:05.601 -> 📶 Deauth: 30:CB:C7:3E:A8:D0 → Unknown Network CH:3 RSSI:-52 Rate:33
16:50:06.181 -> 📶 Deauth: 30:CB:C7:3E:A8:D0 → Unknown Network CH:5 RSSI:-51 Rate:34
16:50:06.418 -> 📶 Deauth: 30:CB:C7:3E:A8:D0 → Unknown Network CH:6 RSSI:-52 Rate:40
16:50:06.612 -> 📶 Deauth: 30:CB:C7:3E:A8:D0 → Unknown Network CH:7 RSSI:-51 Rate:45
16:50:07.140 -> 📶 Deauth: 30:CB:C7:3E:A8:D0 → Unknown Network CH:9 RSSI:-50 Rate:46
16:50:07.426 -> 📶 Deauth: 30:CB:C7:3E:A8:D0 → Unknown Network CH:10 RSSI:-51 Rate:52
16:50:08.139 -> 📶 Deauth: 30:CB:C7:3E:A8:D0 → Unknown Network CH:13 RSSI:-53 Rate:58
16:50:08.328 -> 📶 Deauth: 30:CB:C7:3E:A8:D0 → Unknown Network CH:1 RSSI:-52 Rate:62
16:50:08.518 -> 📶 Deauth: 30:CB:C7:3E:A8:D0 → Unknown Network CH:2 RSSI:-52 Rate:66

```


Figure 2 check the status in the Arduino like a web

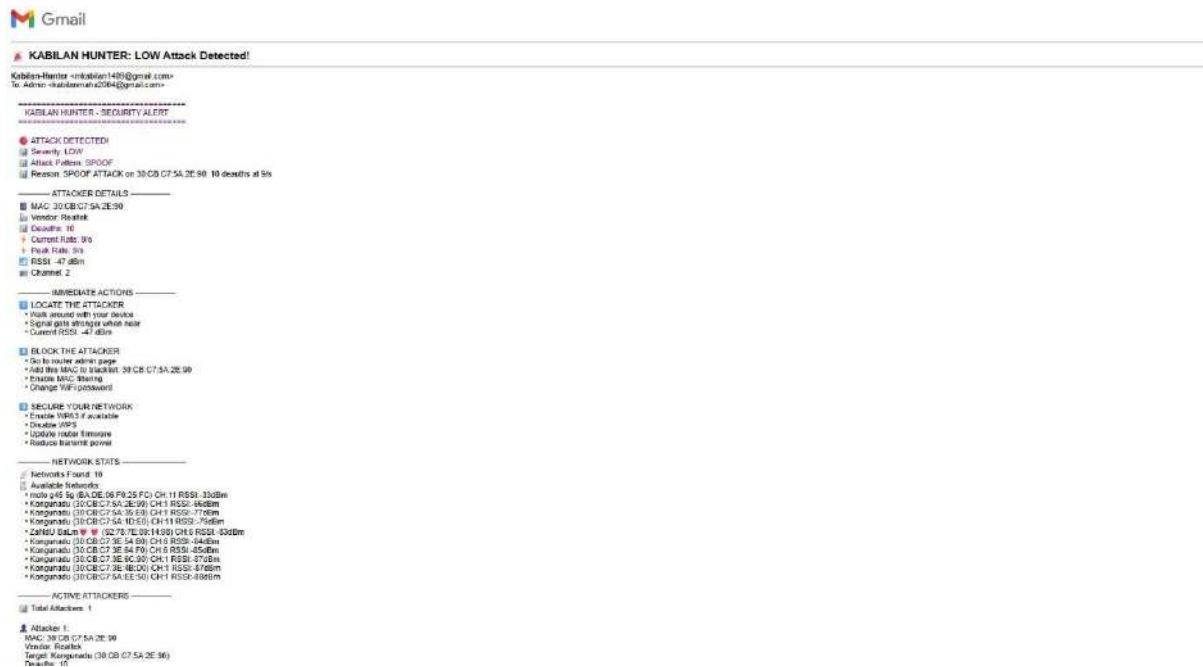


Figure3 alert from the esp32 in the email

6. DISCUSSION

The hybrid detection strategy improves reliability compared to single-metric approaches. Rate-based detection captures burst attacks, count-based detection identifies sustained floods, and spoof detection identifies identity-based attacks.

RSSI filtering significantly reduces false positives from distant sources. Strong RSSI values also allow rough attacker proximity estimation.

The system demonstrates that embedded devices can perform effective intrusion detection without centralized systems.

7. CONCLUSION

This paper presented an ESP32-based area-wide deauthentication detection system for open Wi-Fi environments. By combining multi-channel monitoring, hybrid detection thresholds, spoof identification, and RSSI filtering, the system achieved reliable real-world performance.

The solution is low-cost, portable, and scalable, making it suitable for academic and public network protection.

Future work includes:

- Adaptive thresholding using machine learning
- Multi-node triangulation
- Cloud-based analytics
- Automated mitigation

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